Reminders

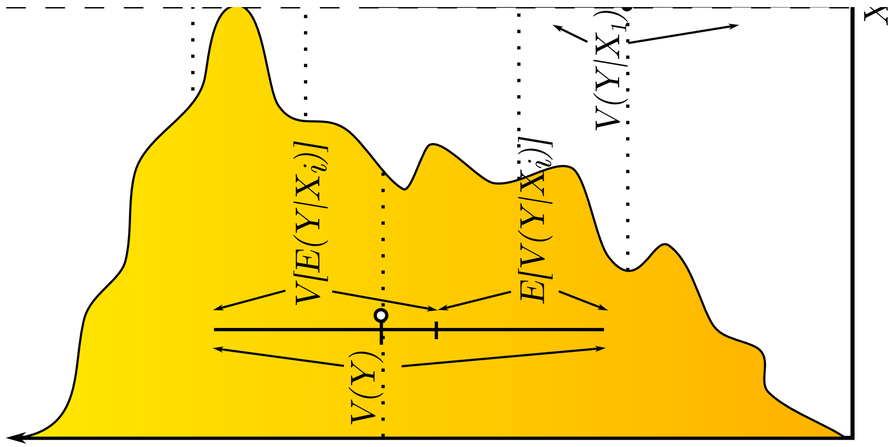
Upcoming due dates
Wed Project Proposal
Fri D4

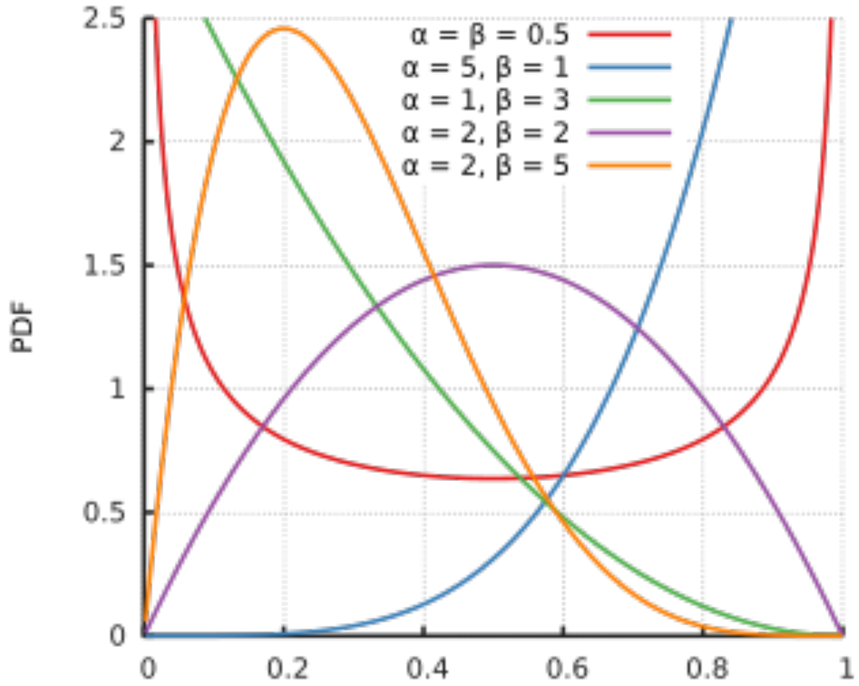
Statistical inference III

Data Science in Practice

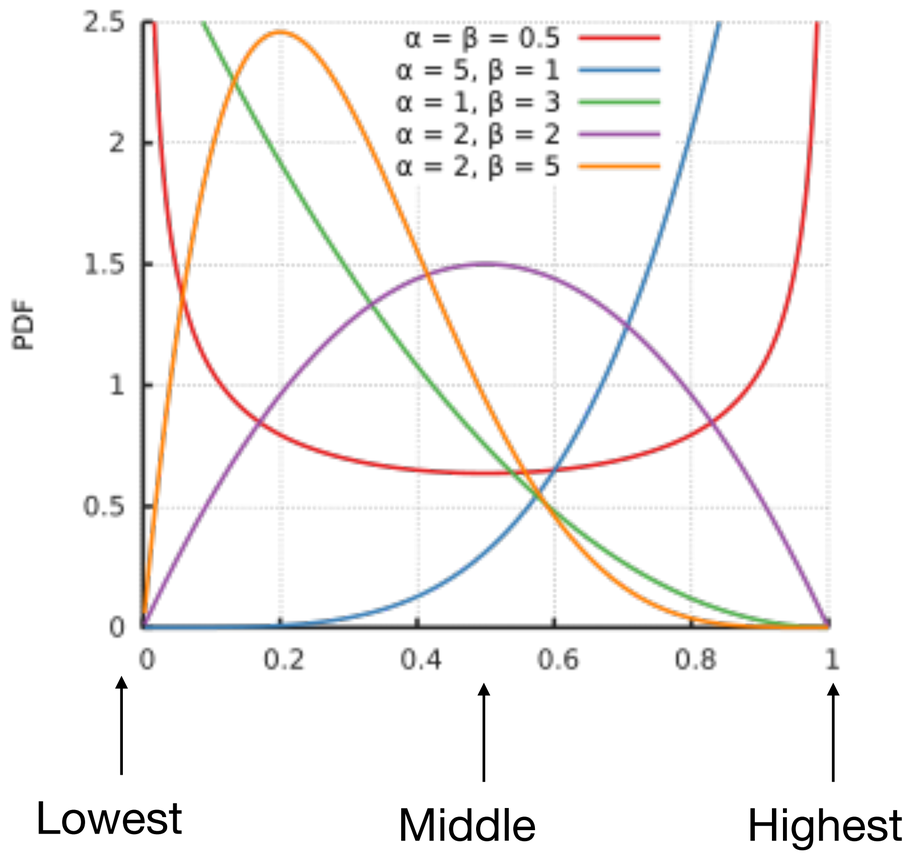
Jason G. Fleischer, PhD Dept. of Cognitive Science UC San Diego https://jgfleischer.com

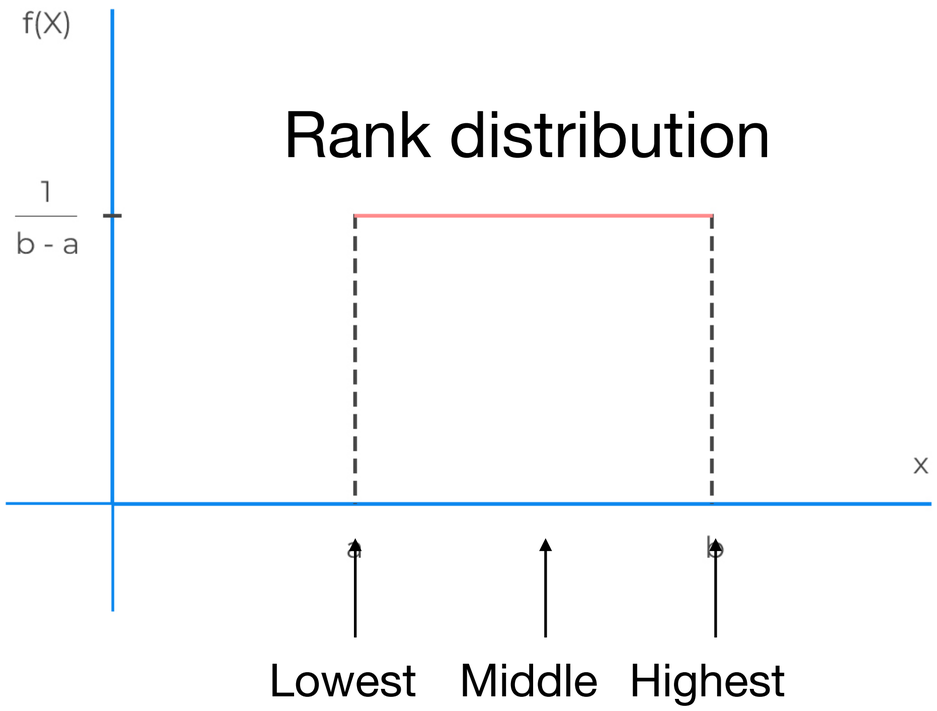
Non-parametric Statistics: What if your distribution looks like this?





Non-parametric Statistics: What if your distribution looks like this? Then rank transform!





Rank Statistics

quantitative data ordinal data

1, 4.5, 6.6, 9.2 → 1, 2, 3, 4

Data are transformed from their quantitative value to their rank.

Ordinal data - categorical, where the variables have a natural order

Particularly helpful when data have a ranking but no clear numerical interpretation (i.e. movie reviews)

Ordinality

Which of the following variables contains ordinal data?

https://forms.gle/UUbbQd9nyz8tgzVh6

Wilcoxon rank-sum test (Mann Whitney U test)

- Determine whether two independent samples were selected from the same populations, having the same distribution
- Similar to t-test (but does not require normal distributions) & tests <u>median</u>

•Assumptions:

- Observations in each group are independent of one another
- Responses are ordinal

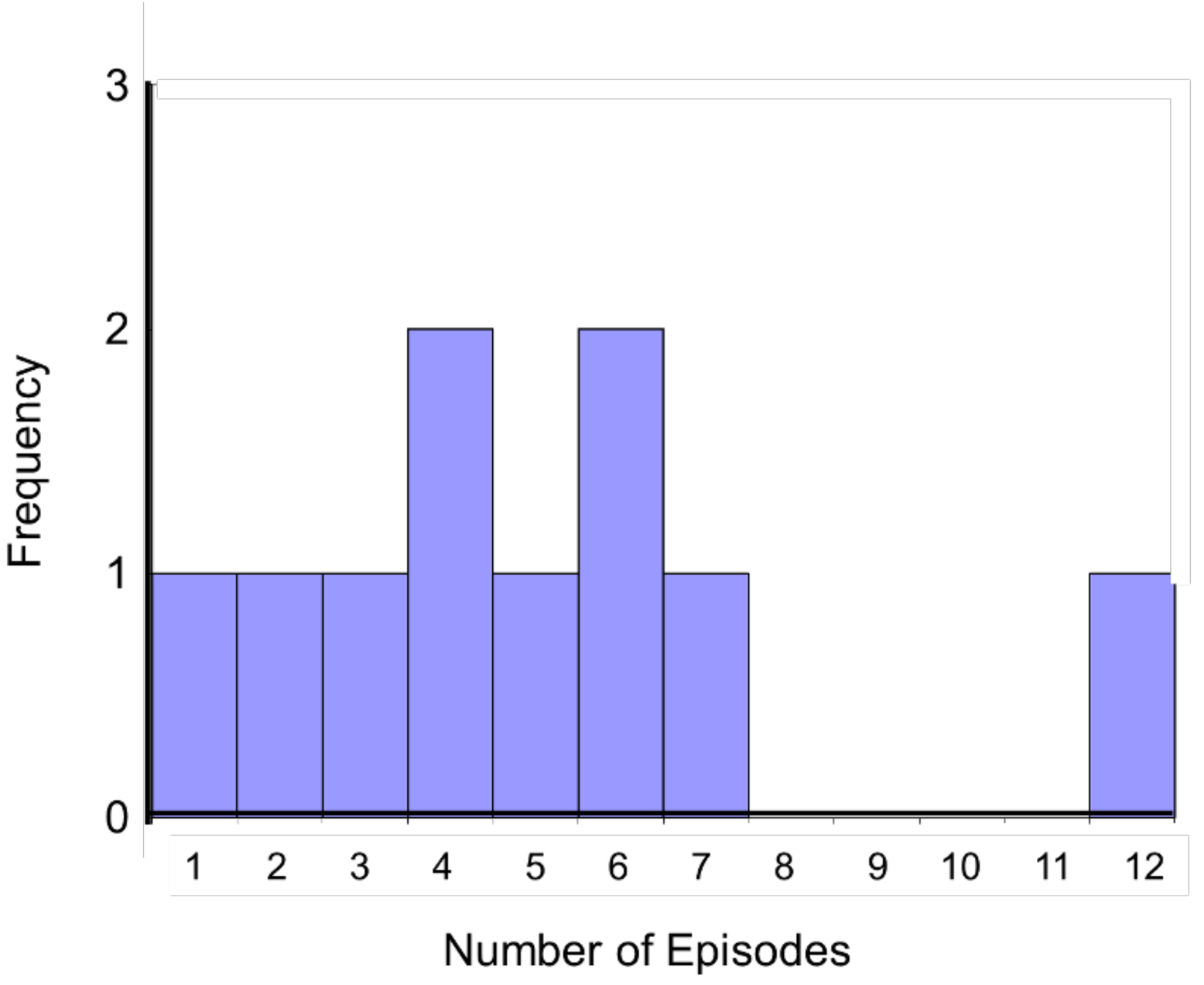
•H_o: distributions of both populations are equal

•H_a: distributions are *not* equal

Mann-Whitney U: question example

- In a clinical trial, is there a difference in the number of episodes of shortness of breath between placebo and treatment?

- Step 1: Participants record number of episodes they have.
- Step 2: Episodes from both groups are combined, sorted, and ranked
- Step 2: Resort the ranks into separate samples (placebo vs. treatment)
- Step 3: Carry out statistical test



http://sphweb.bumc.bu.edu/otlt/mph-modules/bs/bs704_nonparametric/BS704_Nonparametric4.html

		Total Sample (Ordered Smallest to Largest)	Ranks
Placebo	New Drug		
7	3		
5	6		
6	4		
4	2		
12	1		

<u>Sum of ranks:</u>
Placebo = 37
New Drug = 18

http://sphweb.bumc.bu.edu/otlt/mph-modules/bs/bs704_

Mann-Whitney *U*

Ho: low and high scores are approximately evenly distributed in the two groups

Ha: low and high scores are NOT evenly distributed in the two groups (U <= 2)

$$U_A = \boxed{n_a n_b + \frac{n_a(n_a+1)}{2}} - \boxed{T_A}$$

The max possible value of TA

The observed sum of ranks for sample A

n_a = number of elements in group A

n_b = number of elements in group B

$U_{Placebo} = 3$

$U_{treatment} = 22$

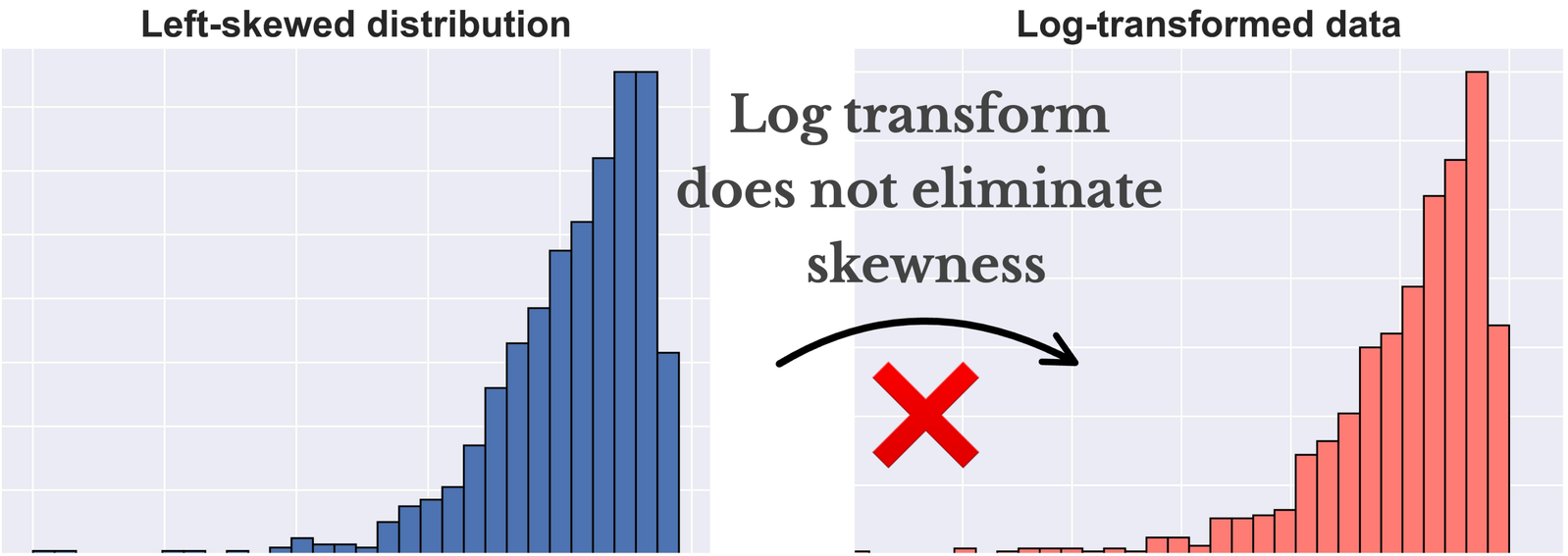
$0 < U < n_1 * n_2$

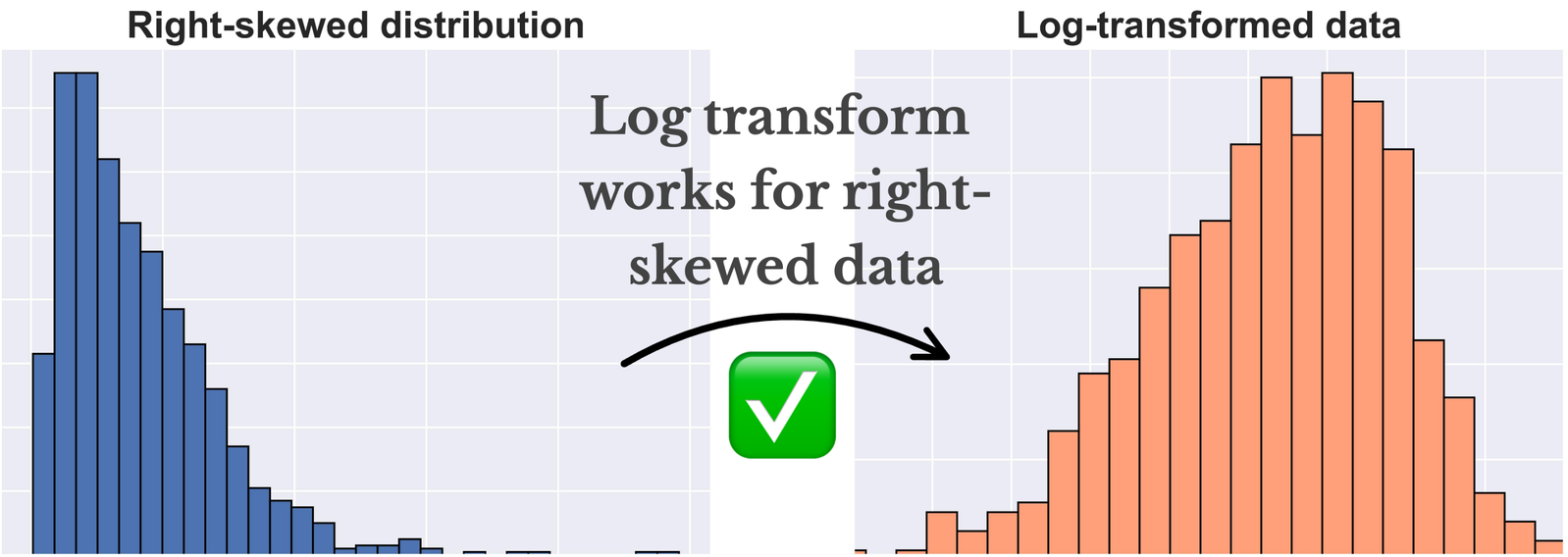
Complete separation → no separation

We reject the null if U is small.

Anything can be transformed!

- Rank transforms are useful with any data which has ordinality!
- Why don’t we use rank based statistics all the time, instead of wasting time with Student’s t etc??
  - We lost statistical power by doing the transform and also from the lack of strong distributional assumptions in the test
- Other transforms can force normality on something non-normal and then we can use normal distribution assumptions and tests. Like, e.g. the log transform…



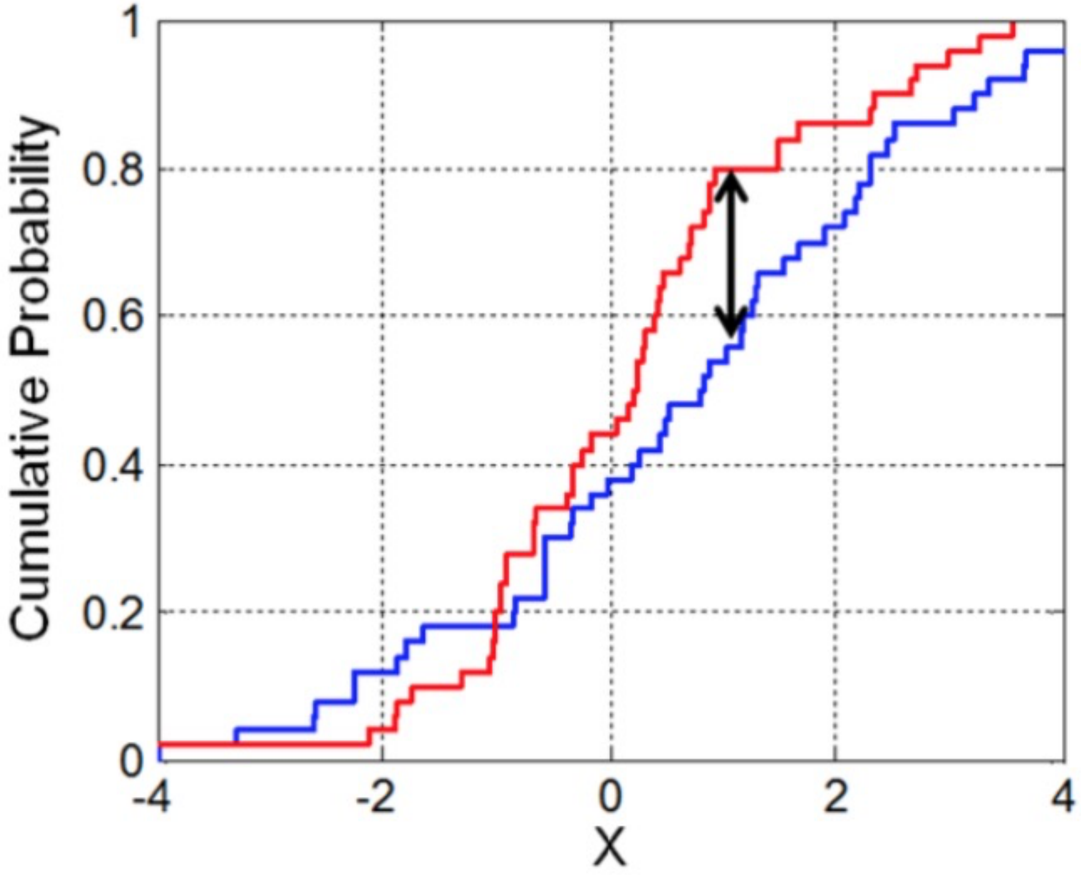



```
log_data = np.log(data+1)
```

Kolmogorov-Smirnov (KS) test

Comparing cumulative distributions regardless of their form

Find the maximum difference between the CDFs.



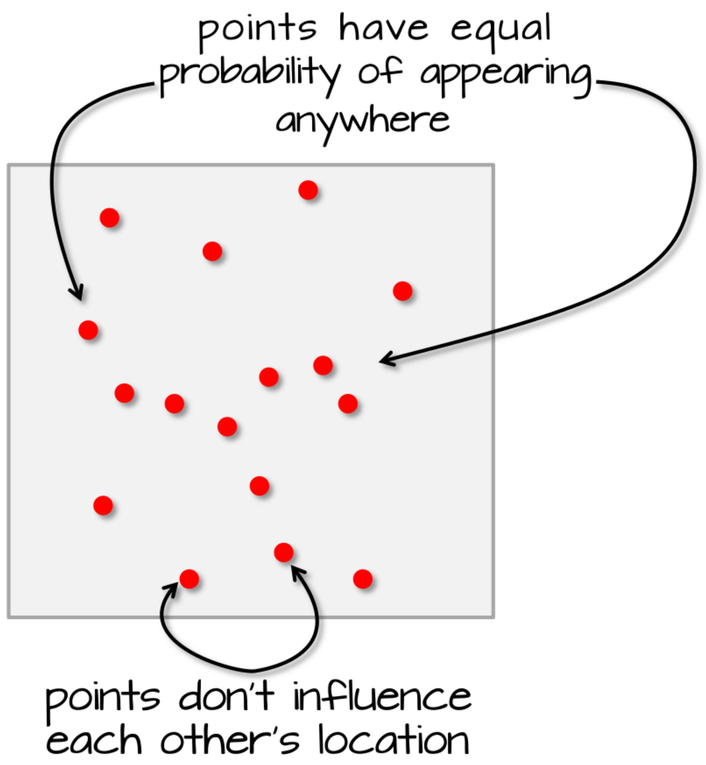
Tests:

- whether a sample is drawn from a given distribution
- Whether two samples are drawn from the same distribution

Hypothesis Testing in Geospatial applications: CSR/IPR

(Distance-based Methods - how the points are distributed relative to one another)

Compare observed point patterns to ones generated by an independent random process (IRP), aka complete spatial randomness (CSR).



CSR/IRP satisfy two conditions:

1. Any event has equal probability of being in any location, a 1st order effect.
2. The location of one event is independent of the location of another event, a 2nd order effect

https://mgimond.github.io/Spatial/introGIS.html

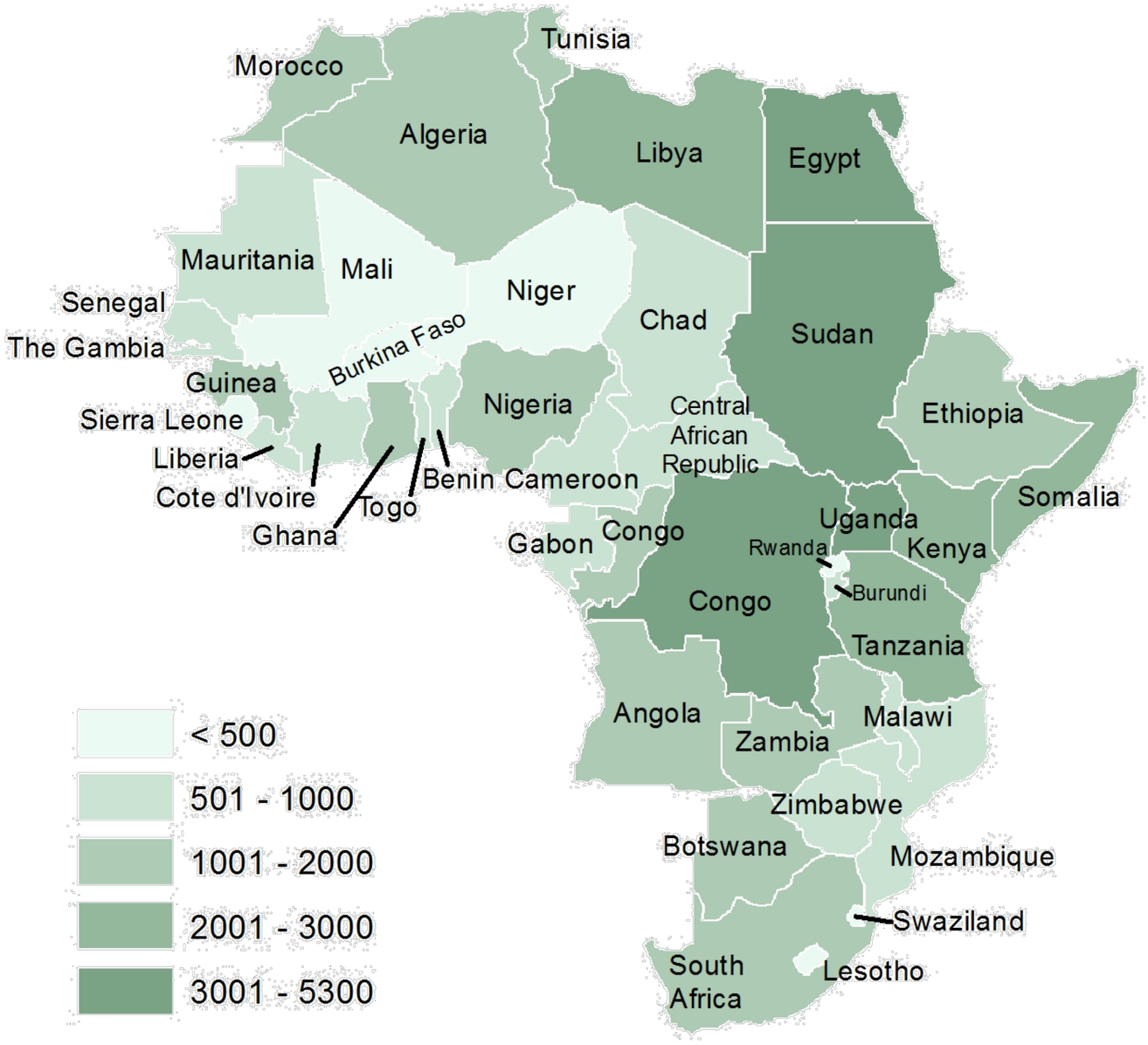
What inference methods are used?

- Monte Carlo simulation testing against a null hypothesis of Complete Spatial Randomness

What metrics are used during inference?

- Density
- Distance to nearest neighbor

Are countries with a high conflict index score geographically clustered?



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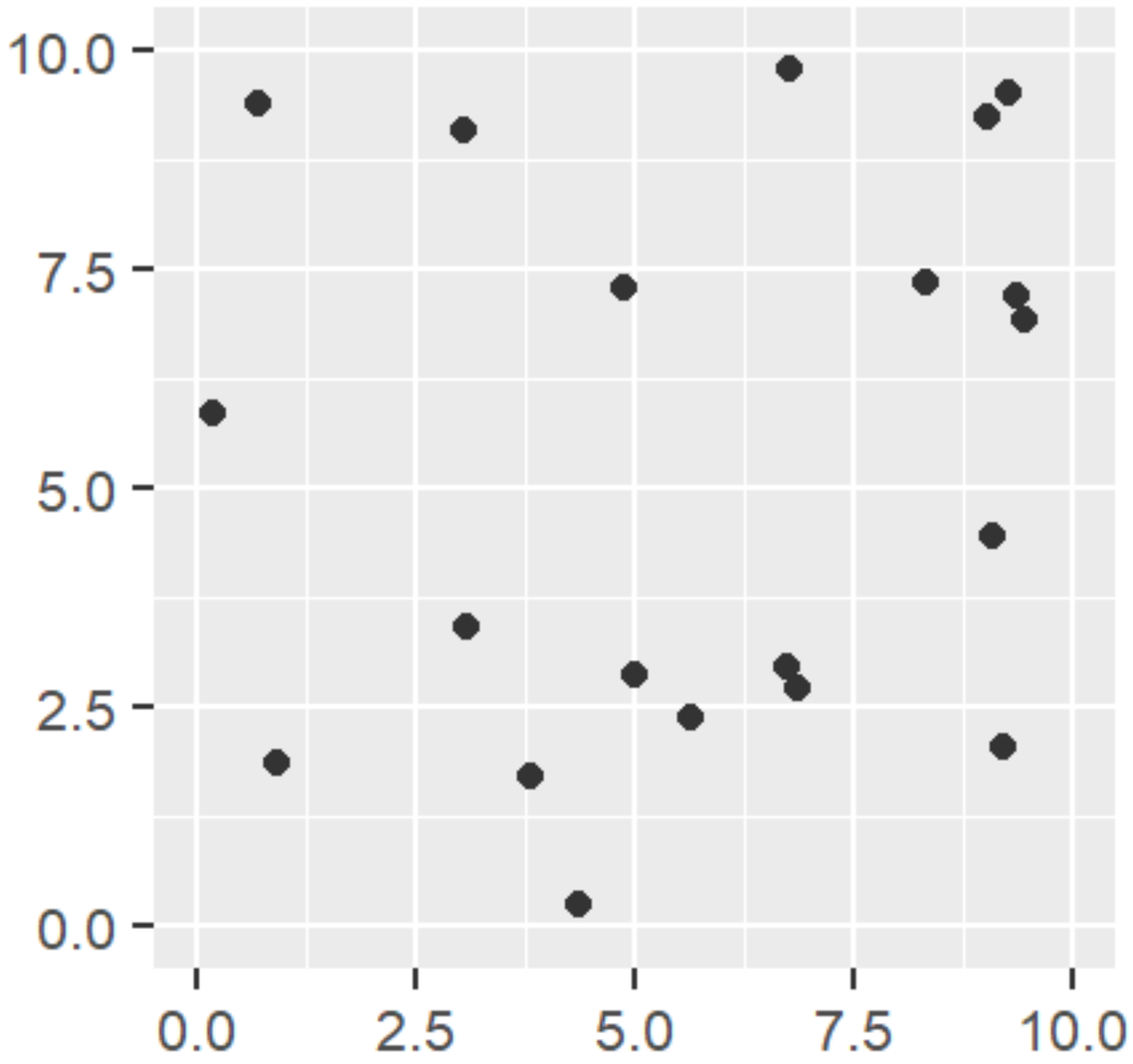
Global Point Density

the ratio of observed number of points to the study region's surface area



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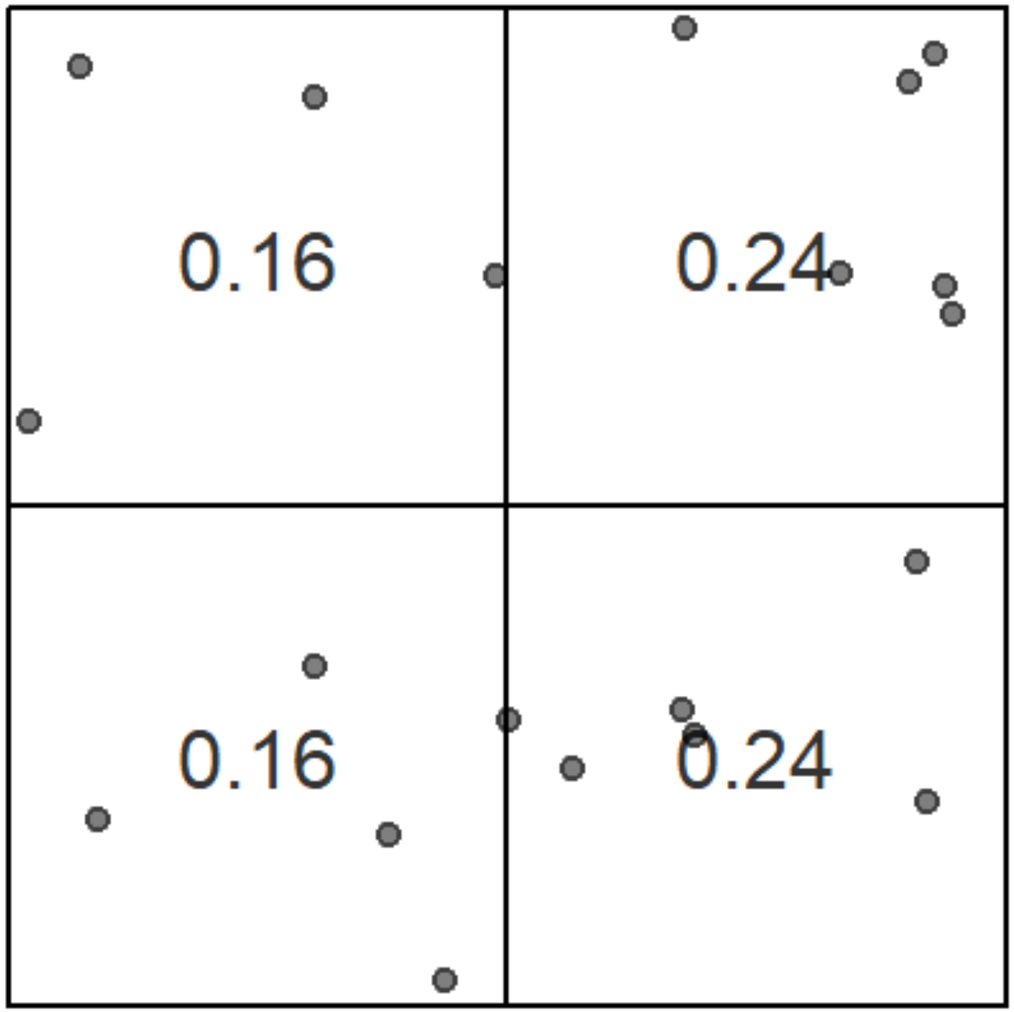
Quadrat Density (local)

Surface is divided and then point density is calculated within quadrat

Note: quadrat number and shape will affect measurement estimate. Suffers from MAUP.



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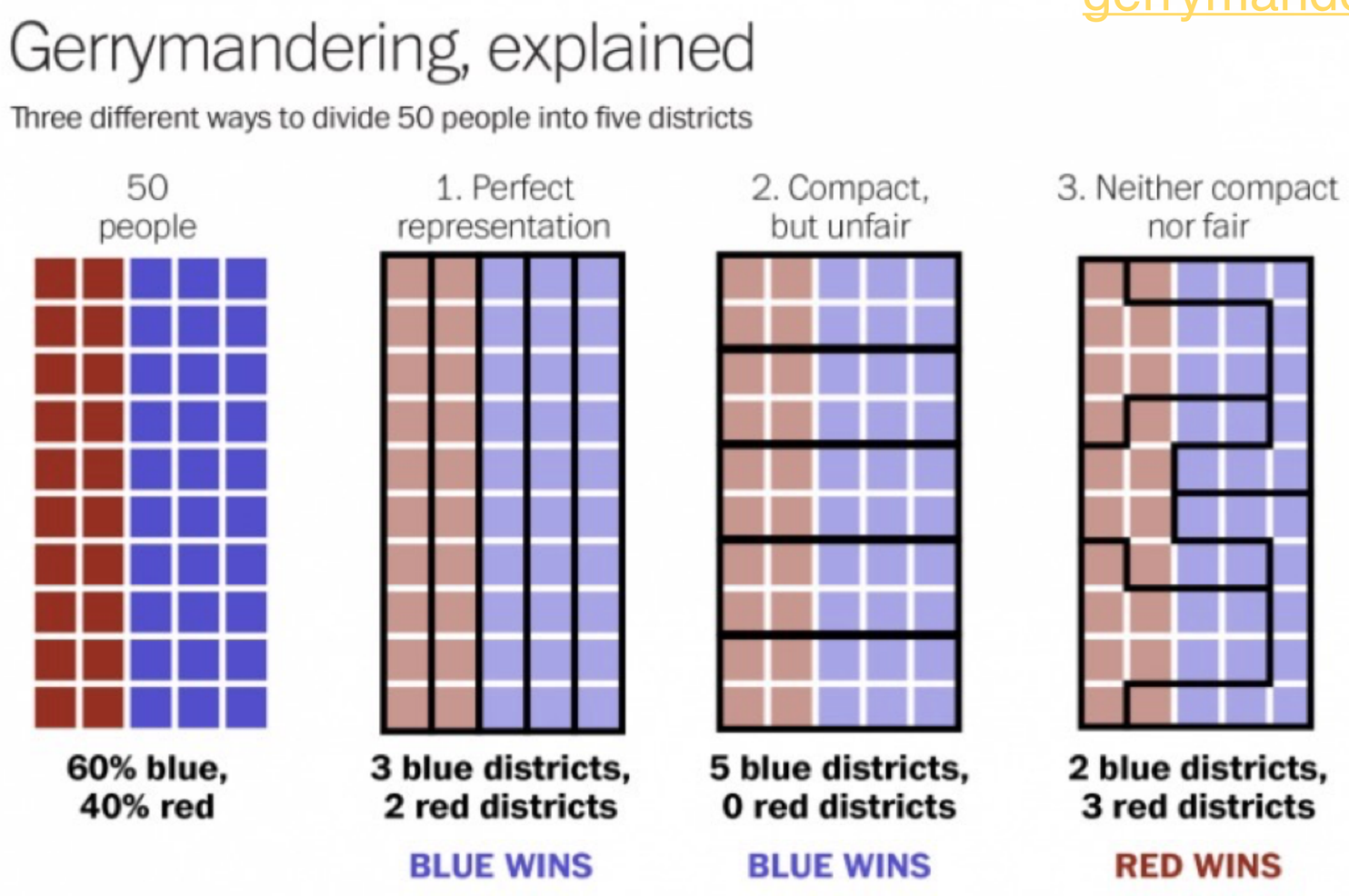
Modifiable Areal Unit Problem (MAUP)

The aggregation units used are arbitrary with respect to the phenomena under investigation, yet the aggregation units used will affect statistics determined on the basis of data reported in this way.

If the spatial units in a particular study were specified differently, we might observe very different patterns and relationships.

For example..

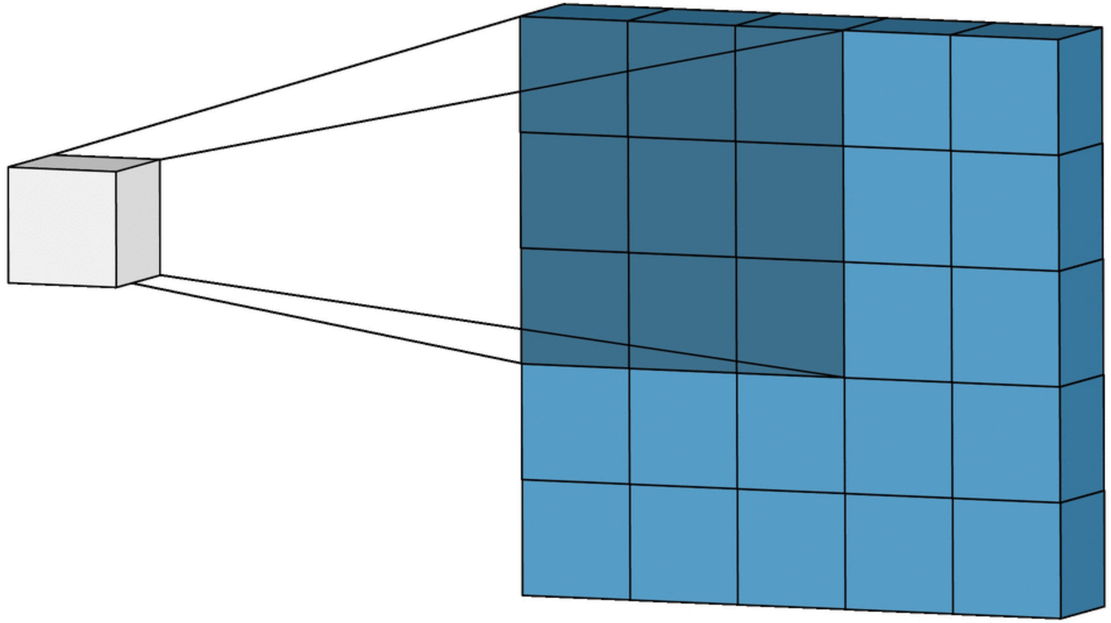
Play this web game to better understand gerrymandering https://www.nytimes.com/interactive/2022/01/27/us/politics/congressional-gerrymandering-redistricting-game-2022.html



Kernel Density (local)

Point density is calculated within sliding windows (window size = kernel)

Note: kernel will affect measurement estimate, but this is less susceptible to MAUP.



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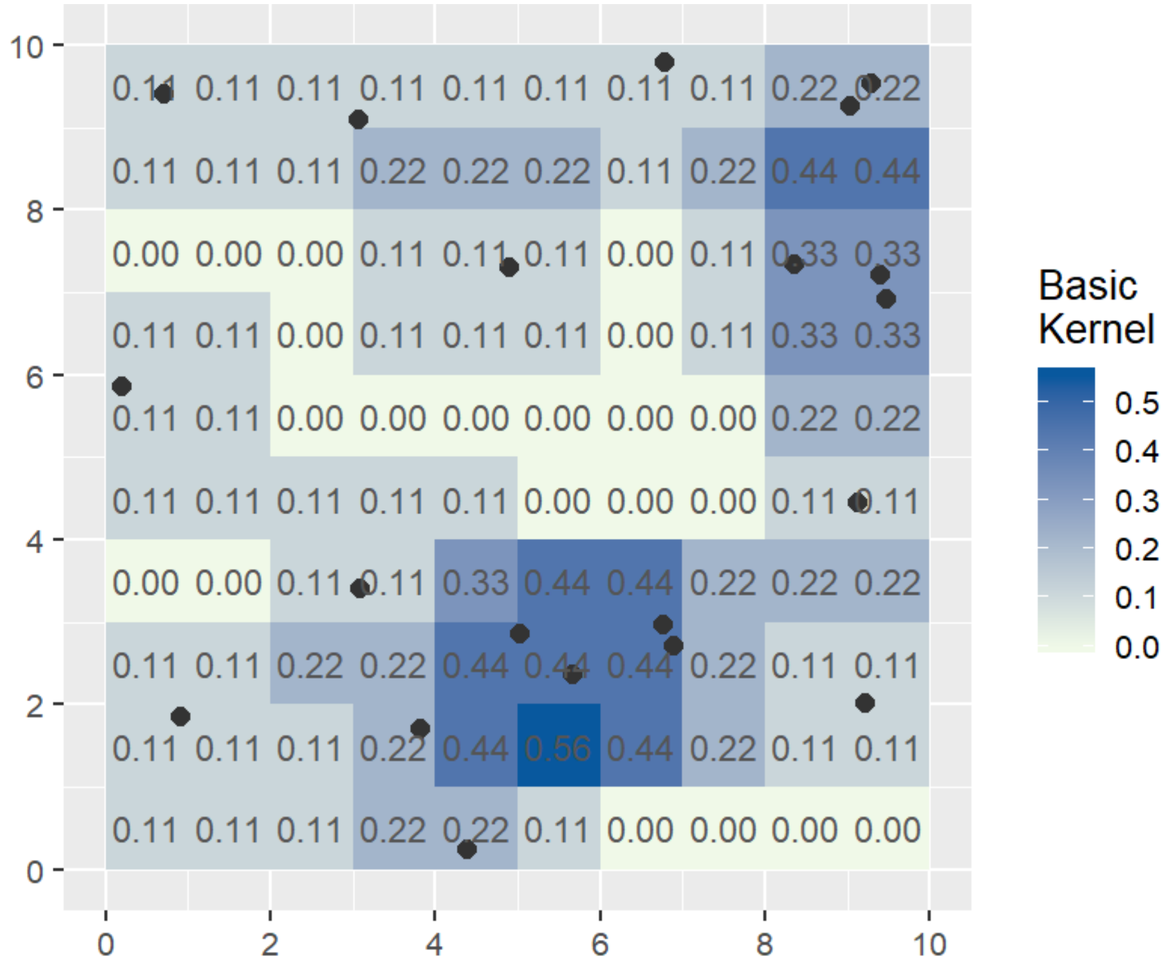

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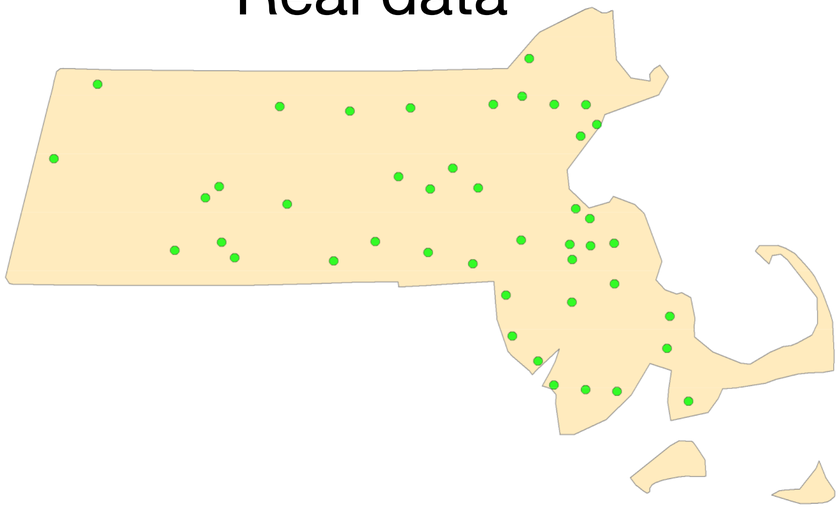
Is this distribution of Walmarts in MA the result of CSR?

Real data



Simulated CSR data

Monte Carlo Simulation

H_o: Distributed randomly (yes CSR)

H_a: NOT distributed randomly (no CSR)

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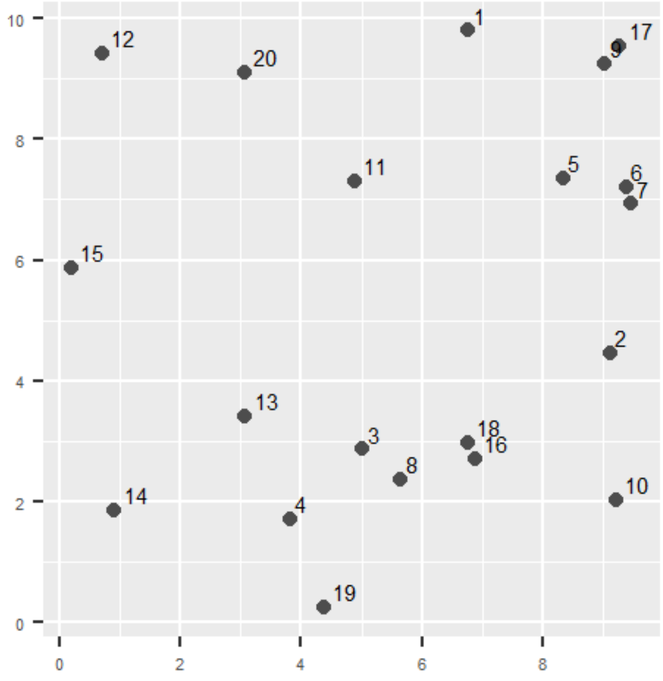
Modeling these data: Average Nearest Neighbor

(Distance-based Methods - how the points are distributed relative to one another)



From	To	Distance	From	To	Distance
1	9	2.32	11	20	2.55
2	10	2.43	12	20	2.39
3	8	0.81	13	4	1.85
4	19	1.56	14	13	2.67
5	6	1.05	15	12	3.58
6	7	0.3	16	18	0.29
7	6	0.3	17	9	0.37
8	3	0.81	18	16	0.29
9	17	0.37	19	4	1.56
10	2	2.43	20	12	2.39

ANN = 1.52 units

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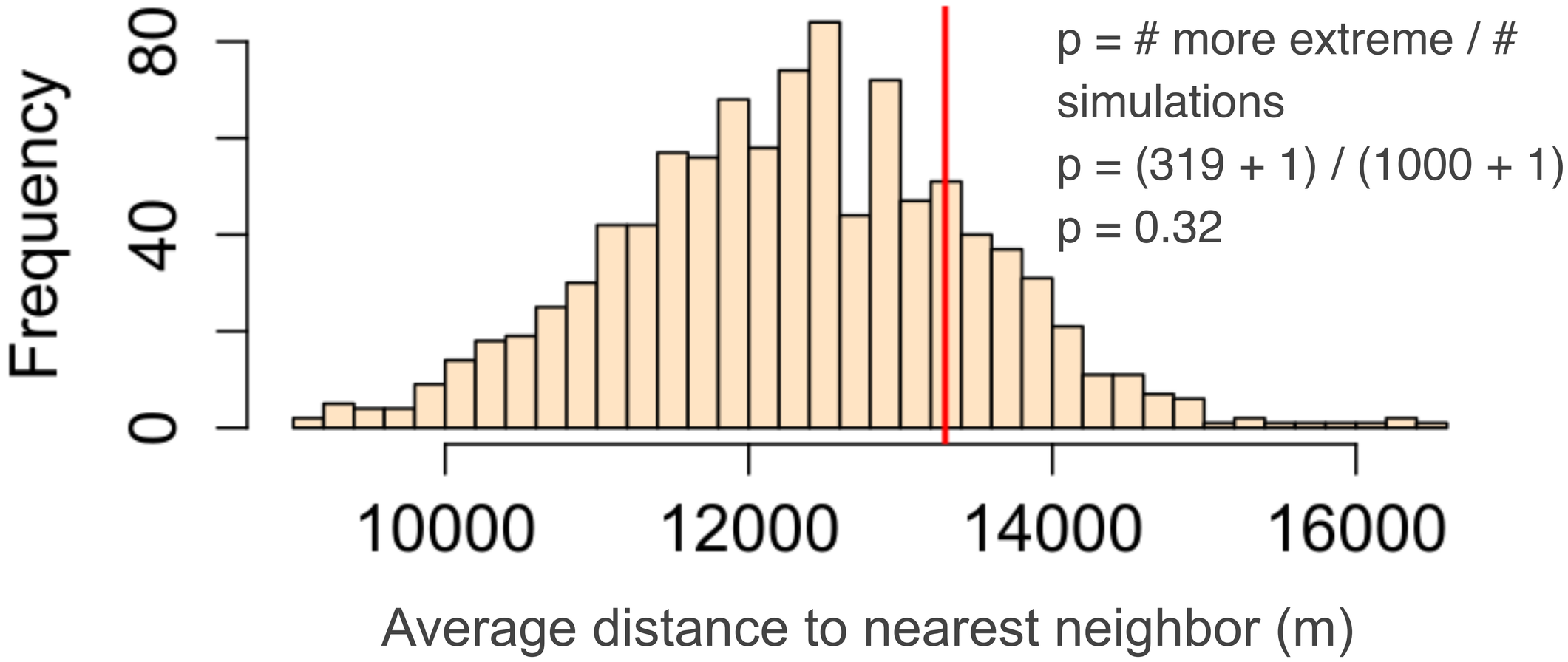

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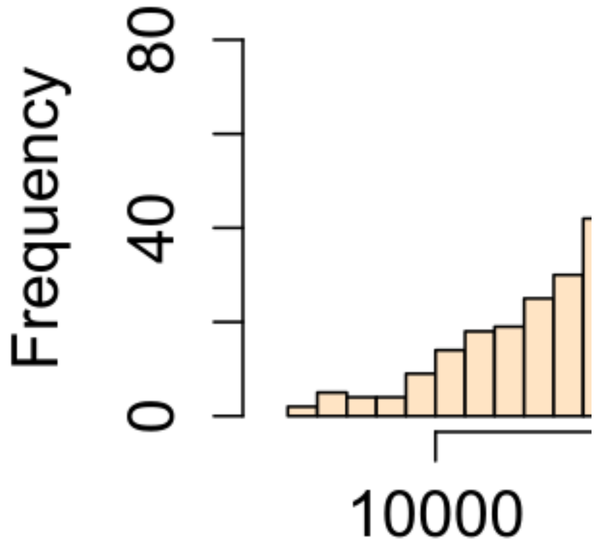


What does the histogr

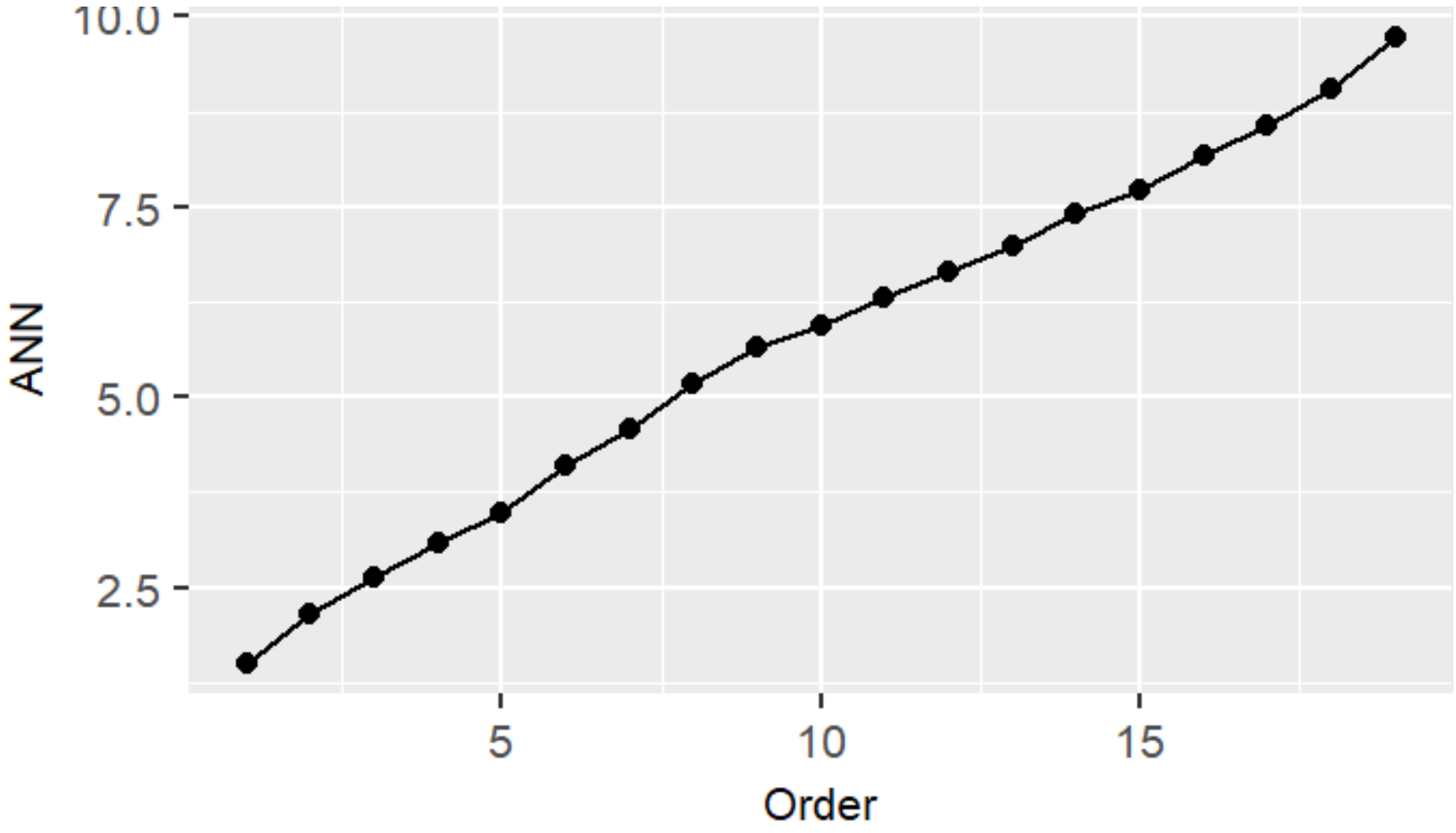
A
Alternative
Hypothesis

B
Null
Hypothesis

https://forms.gle/5a2vacFEz3qLjU7K9

We aren't just limited to "nearest neighbor" and calculating a statistic

plot the ANN values for different order neighbors, that is for the first closest point, then the second closest point, and so forth.



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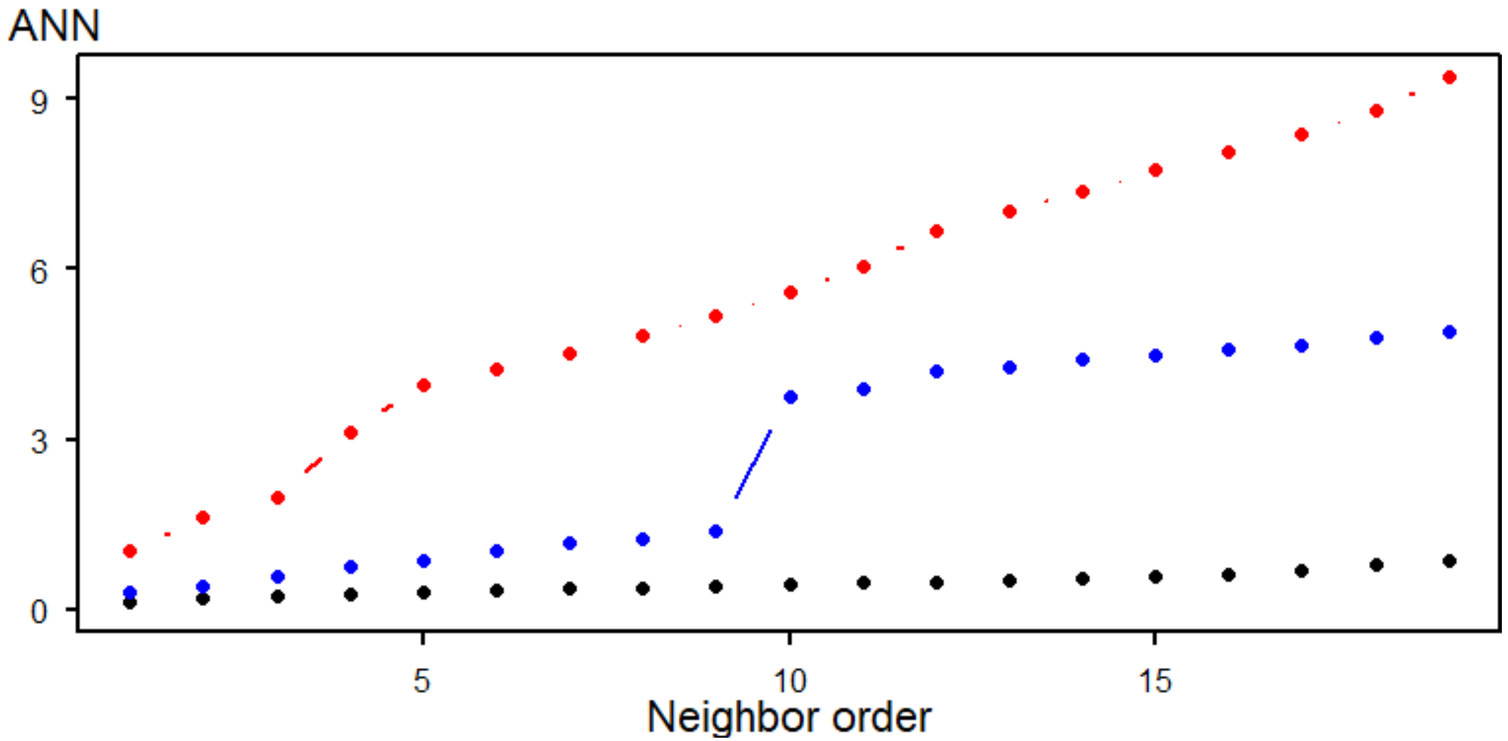
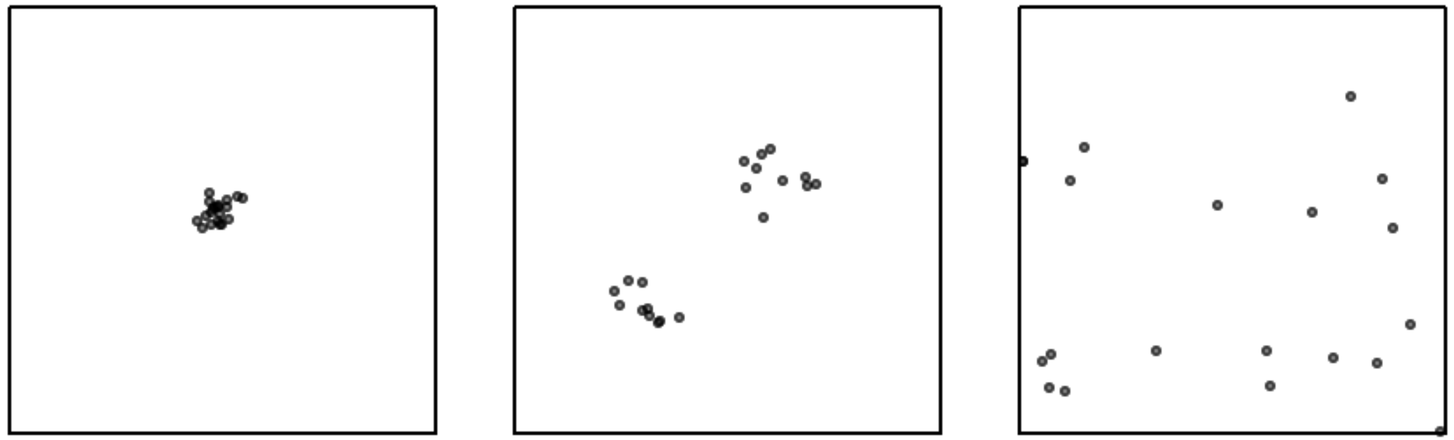
ANN vs neighbor order offers insight into underlying spatial relationship



Note: study space definition affects this measure



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